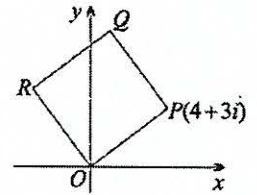


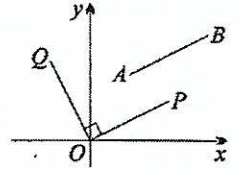
Complex Number 5

1. $OPQR$ is a square. The point P represent $4 + 3i$. Find the complex number represented by

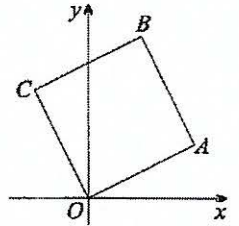
- The point R
- The point Q
- The vector QR
- The vector PR



2. In the diagram, intervals AB , OP and OQ are equal in length, OP is parallel to AB and $\angle POQ = \frac{\pi}{2}$. If A and B represent the complex number $3 + 5i$ and $9 + 8i$ respectively, find the complex number which is represented by Q .

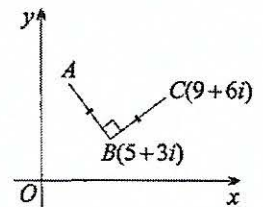


3. $OABC$ is a square. The point A represents the complex number $2 + i$.
- Find the number represented by B and C .
 - If the square is rotated 45° anticlockwise about O to give $OA'B'C'$, find the number represented by B' .



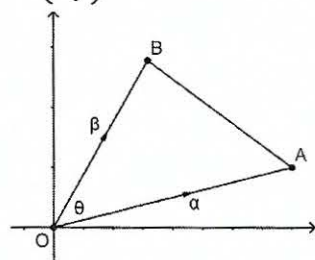
4. In the diagram on the right, $AB = BC$ and $\angle ABC = 90^\circ$. The point B and C represent $5 + 3i$ and $9 + 6i$ respectively. Find the complex number represented by:

- The vector BC
- The vector BA
- The point A .



5. A and B are two points in an Argand diagram representing the complex numbers $z_1 = -1$ and $z_2 = \cos \theta + i \sin \theta$ respectively where $\frac{\pi}{2} < \theta < \pi$. C is the point representing the complex number $z_3 = z_1 + z_2$
- Sketch the quadrilateral $OACB$ on an Argand diagram where O is the point representing the complex 0. Then mark an angle on the diagram which is equal to θ .
 - Let $z_4 = z_2 - z_1$
 - Show that $\frac{z_4}{z_3} = i \left(\frac{\sin \theta}{\cos \theta - 1} \right)$,
 - Using part i. to show that the diagonals of the quadrilateral $OACB$ are perpendicular to each other.
6. If $z = \cos \theta + i \sin \theta$:
- Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$
 - Let $\omega = z + \frac{1}{z}$. Prove that $\omega^3 + \omega^2 - 2\omega - 2 = z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3}$.
 - Hence solve $\cos \theta + \cos 2\theta + \cos 3\theta = 0$ for $0 \leq \theta \leq 2\pi$
7. Given ω is a complex root of $z^3 - 1 = 0$
- Explain carefully why ω^2 is also a complex root of $z^3 - 1 = 0$.
 - Prove that $1 + \omega + \omega^2 = 0$
 - Show the roots of $z^3 - 1 = 0$ on an Argand diagram.
 - Simplify $(1 + \omega^2)^{2012}$ completely.
8. a. Given that ω is a complex cube root of unity, show that ω^2 is the other complex root.
- Plot the three cube roots of unity on an Argand diagram.
 - Show that $1 + \omega + \omega^2 = 0$.
 - Simplify $(7 + 9\omega^4 + 7\omega^{-1})^6$

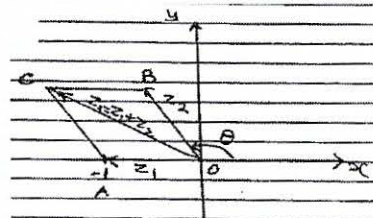
9. Let $z_2 = 1 + i$ and, for $n > 2$, and $z_n = z_{n-1} \left(1 + \frac{i}{|z_{n-1}|}\right)$. Use mathematical induction to prove that $|z_n| = \sqrt{n}$ for all integers $n \geq 2$.
10. Describe geometrically the transformation $z \rightarrow \alpha z$, where $\alpha = -2 + 2i$. Illustrate on an Argand diagram for $z = 3i$.
11. a. If p is real, and $-2 < p < 2$, show that the roots of the equation $x^2 + px + 1 = 0$ are non-real complex numbers with modulus 1.
 b. Solve the equations $x^2 + x + 1 = 0$ and $x^2 - \sqrt{3}x + 1 = 0$. Plot on an Argand diagram the points A and B representing the solutions of the first equation, and C and D representing the solutions of the second, choosing A and C to lie above the real axis.
 c. Find the angles $\widehat{AOB}$, $\widehat{COD}$, $\widehat{COA}$ and $\widehat{ACB}$.
12. a. Obtain in the form $a + ib$ the roots of the equation $x^2 + 2x + 3 = 0$. Find the modulus and argument of each root and represent the roots on an Argand diagram by the points A and B .
 b. Let H and K be the points representing the roots of $x^2 + 2px + q = 0$, where p and q are real and $p^2 < q$. Find the algebraic relation satisfied by p and q in each of the following cases
 i. $\widehat{HOK}$ is a right angle.
 ii. A, B, H and K are equidistant from O
13. On an Argand diagram the points A and B represent the complex numbers $z_1 = i$ and $z_2 = \frac{1}{\sqrt{2}}(1 + i)$ respectively. Show that $\arg(z_1 + z_2) = \frac{3\pi}{8}$.
14. a. Use the results $z + \bar{z} = 2\operatorname{Re}(z)$ and $|z|^2 = z\bar{z}$ to show that $|\alpha|^2 + |\beta|^2 - |\alpha - \beta|^2 = 2\operatorname{Re}(\alpha\bar{\beta})$.
 b. The diagram shows the angle θ between the complex numbers α and β . Prove that $|\alpha\beta| \cos \theta = \operatorname{Re}(\alpha\bar{\beta})$



15. Given $1 - 2i$ is one root of the equation $x^2 + (1 + i)x + k = 0$, find the other root and the value of k .
16. a. If $|z| = 1$, show that $z^n - z^{-n} = 2i \sin n\theta$
 b. Hence show that $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$
17. Draw neat sketches of: $0 < \arg[(1 - i)z] < \frac{\pi}{6}$

Answer:

1. a. $-3 + 4i$ b. $1 + 7i$ c. $-4 - 3i$ d. $-7 + i$
 2. $-3 + 6i$
 3. a. B represents $1 + 3i$, C represents $-1 + 2i$ b. $-\sqrt{2} + 2\sqrt{2}i$
 4. a. $4 + 3i$ b. $-3 + 4i$ c. $2 + 7i$
 5. a.



6. c. $\theta = \frac{\pi}{4}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{4}$

7. d. ω^2

8. d. 64

10. Rotation anticlockwise by $\frac{3\pi}{4}$, enlargement by $2\sqrt{2}$; $3i \rightarrow -6(1 + i)$

11. $x = -\frac{1}{2}(1 \pm \sqrt{3}i); \frac{2\pi}{3}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{\pi}{3}$

12. $x = -1 \pm \sqrt{2}i, \sqrt{3}cis[\pm(\pi - \tan^{-1}\sqrt{2})]; 2p^2 - q = 0, q = 3$

15. $-2 + i, k = 5i$

17. $\frac{\pi}{4} < \arg z < \frac{5\pi}{12}$

